

Multiple Choice

1. **Answer:** C
Options: A) 1; B) 2; C) 2,3; D) 6
Note: Correct option: C - 2,3
2. **Answer:** D
Options: A) 1; B) 2; C) 3; D) 4
Note: Correct option: D - 4
3. **Answer:** B
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: B - False, False
4. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
5. **Answer:** B
Options: A) the entire real axis; B) a point; C) a ray; D) an open finite interval
Note: Correct option: B - a point

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Yes, with $p=3$ '.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'False, True'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** True, True. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** I and II only. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.